

Reviewer Pack — Minimal Order of Hypergeometric Functions Bounds Circuit Siz...

Entry 8c38c25676b5 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Hypergeometric Functions Bounds Circuit Size of k-CNF

Entry ID: 8c38c25676b5 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-29 22:42:49 UTC

1. Conjecture

Field A (mathematical branch): Hypergeometric Functions **Field B** (complexity object): Boolean Function Complexity

Statement:

For a given k , the minimum order of a hypergeometric function that uniquely represents each possible assignment of k variables in a k -CNF formula is $\Theta(n^{(k/2)})$ for $n \leq 40$.

Rationale (proposer's reasoning):

Hypergeometric functions are known to be expressive in capturing polynomial relationships, and their minimal orders could potentially reveal structural properties of boolean circuits. If the conjecture holds, it would suggest a deep connection between hypergeometric function theory and circuit complexity, potentially providing new insights into proving lower bounds for k -CNF satisfiability.

Taxonomy category: cg_hypergeo (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 594fde30a4a550a1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

If the mean order of hypergeometric functions across all k -CNF assignments is $\leq n^{(k/2)}$ for all $n \leq 40$, and this mean is supported by at least 80% of the seeds, or if any seed produces an order $> n^{(k/2)}$, then the conjecture is falsified.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 2 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "hypergeometric functions" AND "circuit size" AND k-CNF - "Boolean function complexity" AND "minimum order" AND hypergeometric - " $\Theta(n^{(k/2)})$ " AND hypergeometric functions AND Boolean function complexity

Top relevant hits considered: - [s2:10.1515/dma-2021-0031] On the complexity of monotone circuits for threshold symmetric Boolean functions - [s2:e7576d204ed2ef40f290bc8b3444597fdfea5cf4] Domain Reduction for Monotonicity Testing: A $o(d)$ Tester for Boolean Functions on Hypergrids

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, $\leq 90s$ wall time per cycle **Execution:** rc=1, elapsed=1.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def hypergeometric_order(n, k):
        if n == 0 or k == 0:
            return 1
        order = 1
        for i in range(1, k + 1):
            order *= (n - i + 1) / i
        return int(math.ceil(order))

    def generate_k_cnf(k, n):
        cnf = []
        for _ in range(n):
            clause = [random.choice([-1, 1]) * random.randint(1, k) for _ in range(k)]
            cnf.append(clause)
        return cnf

    def calculate_order(cnf):
        orders = []
        for clause in cnf:
            n = len(clause)
            order = hypergeometric_order(n, k)
            orders.append(order)
        return orders

    k_values = [5, 10, 15, 20, 30, 40]
```

```

total_orders = []
instances_tested = 0
n_max = 0

for k in k_values:
    cnf = generate_k_cnf(k, 10) # Generate 10 instances per k
    orders = calculate_order(cnf)
    total_orders.extend(orders)
    instances_tested += len(orders)
    n_max = max(n_max, k)

mean_order = sum(total_orders) / instances_tested
conjecture_holds = all(order <= n**(k/2) for order in total_orders)
counterexample = "" if conjecture_holds else "mapping_undefined"

return {
    "metric_name": "order",
    "metric_value": mean_order,
    "instances_tested": instances_tested,
    "n_max": n_max,
    "conjecture_holds": conjecture_holds,
    "counterexample": counterexample
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_order = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_order} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_order} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

--STDERR--

Traceback (most recent call last):

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_498f32e3.py", line 75, in <module>
    result = run_trial(seed)

```

```

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_498f32e3.py", line 57, in run_trial
    conjecture_holds = all(order <= n**(k/2) for order in total_orders)
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_498f32e3.py", line 57, in <genexpr>
    conjecture_holds = all(order <= n**(k/2) for order in total_orders)

```

8. Critic adversarial review

Critic reasoning:

9. Final verdict & safety rail

Reasoning:

11. Audit log (LLM calls)

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14135
2	propose	ollama_remote	glm4:latest	0	0	22310
3	preregistration	ollama_remote	glm4:latest	0	0	23933
4	novelty	ollama_remote	glm4:latest	0	0	9615
5	novelty	ollama_remote	glm4:latest	0	0	17557
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16661
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9830
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9080
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	21223
10	judge	ollama_remote	glm4:latest	0	0	17875

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 162218 ms total latency. Provider mix: {'ollama remote': 10}

(full prompt+response transcripts available in [research/audit/8c38c25676b5.jsonl](#))

12. Reproducibility

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/8c38c25676b5.jsonl` for verification of provenance

4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/8c38c25676b5.tar.gz` (if generated)